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Static and Dynamic Embedding Approaches to Identify Is-A Relations in SNOMED CT

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***Abstract*—**

Systematized Nomenclature of Medicine - Clinical Terms (SNOMED CT) is a comprehensive clinical terminology with over 350,000 unique concepts and 1 million relationships. It serves as an essential resource for both clinical practice and medical research. Approximately 40% of these relationships are of the *is-a* type, which is key to classifying concepts as subtypes or subclasses within broader categories and is fundamental for decision support systems and automated reasoning in clinical environments. Identifying *is-a* relationships in SNOMED CT is crucial for enhancing the accuracy of patient cohort queries, which form the backbone of clinical and research applications. Our study aims to use deep learning-based neural networks to determine whether a relationship between two concepts falls under the *is-a* or *non-is-a* categories. This approach can help detect misclassified or undefined *is-a* concept pairs, ultimately improving the quality of SNOMED CT ontology. We construct a node-edge-node structure for concept pairs and their relationships, which we then split into two categories: *is-a* and *non-is-a* classes. In this study, we propose two classification approaches. In the first approach using embeddingBag, the data are mapped into vector representations and used as input for a fully connected neural network to learn the patterns of the *is-a* relationship. During model training, 80% of the data is used for training and the remaining 20% for testing.

We achieved a precision of 0.903, recall of 0.894, and F1 Score of 0.898 in predicting the *is-a* relation among the associated medical concepts. The second approach involved transformer-based models like BERT(Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. [1], which dramatically advanced various natural language processing (NLP) tasks by offering deeply contextualized word embeddings. And RoBERTa(Robustly optimized BERT approach)[2], both BERT and RoBERTa models have significantly advanced the field of NLP. This work uses them as encoders connected to a classifier head to predict *is-a* or *non-is-a* categories. We fine-tune our models using a historical SNOMED CT dataset from 2017, while evaluation is carried out on new data introduced after 2023 in the 2024 release. With a recall of 0.933, precision of 0.933, F1 score of 0.932, accuracy of 0.933, and ROC of 0.972, RoBERTa outperforms the other baseline models (Support Vector Machine(SVM), K-Nearest Neighbors(KNN), Naive Bayes, and Multilayer Perceptron(MLP)) across all metrics. This work demonstrates that our implementation can effectively identify unknown *is-a* relations in future datasets.

***Index Terms*—Classification of Multi Sentence, Deep Learning, Information Retrieval, Natural Language Processing, SNOMED-CT, Large Language Model, BERT**

I. Introduction

Medical ontologies are vocabulary systems that standardize and precisely express medical concepts and their relationships to enhance the effectiveness in communication and exchange of information among medical professionals and reduce misunderstandings[3], [4], [5]. The SNOMED CT is an extensively used medical ontology within the healthcare domain[6], with a comprehensive and hierarchical collection of terms that span various medical fields. As a unified and precise terminology system, SNOMED CT has garnered wide-ranging global adoptions and recognitions within the international medical and health communities[7], [8]. SNOMED CT is essential in facilitating accurate representation and understanding of medical concepts, contributing to enhanced clinical decision-making, research, and interoperability across healthcare systems.

Its hierarchical structure, especially the *is-a* relationship, plays a pivotal role in ensuring the semantic richness of medical data. Gaps in these hierarchies can result in missed diagnoses or inappropriate alerts, directly affecting patient care and safety. For instance, the absence of an *is-a* relationship between "Osteoarthritis of knee" and "Arthritis of knee" could lead to incomplete identification of arthritis-related patient cohorts, thereby compromising the quality and reliability of healthcare research and operational decisions. Therefore, motivated by the critical importance of these relationships, we aim to study the SNOMED CT ontology using deep learning approaches to better analyze and understand the features of *is-a* relationship.

Modeling and analyzing medical ontology has been an active research area with many methods and procedures proposed and studied, including word embedding and converting graph-based data into a text corpus with labeled annotations[11]. Mohtashamian et al. [12] aim to enhance the Orphanet Rare Disease Ontology (ORDO) by identifying missing *is-a* relationship, crucial for accurate data integration and retrieval in biomedical research. Since *is-a* is also a major type of relation in SNOMED CT, we can improve its information accuracy by developing a solution to identify potential missing *is-a* relationships through learning the lexical features of *is-a* related concept descriptions.

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In our study, we first develop a neural network classification model to identify concept pairs with an *is-a* relation, employing the EmbeddingBag technique[13] to address the challenge of normalizing input with varying lengths. The need for labeled data with ground truth for training and evaluation is paramount for classification tasks on a corpus[14]. However, manual labeling of datasets requires significant human efforts[15]. The SNOMED CT dataset initially contains multiple fields, including source and destination identifiers, relation groups, type identifiers, and textual representations of terms and their relationships. We preprocess the dataset to streamline it for binary classification, distinguishing between *is-a* and other *non-is-a* relationship types. To create a dataset for the classification task, we generate positive and negative instances from the SNOMED CT database by categorizing related concept pairs into two groups: one containing *is-a* relationships and the other *non-is-a* group containing pairs linked by relation types other than *is-a*. This approach allows us to obtain accurate ground truth data without the costs associated with manual labeling. Furthermore, our method creates balanced datasets of positive and negative instances, which collectively contribute to generating high-quality training data. We also propose a fusion embedding method to combine the feature representations of both concepts by selecting the highest value from their embeddings before any processing or learning tasks. These techniques collectively lead to our proposed deep learning model, demonstrating high performance in preventing overfitting and enhancing robustness. This model accurately predicts whether two concepts should be connected by an *is-a* relation. Specifically, we focus on predicting potential missing relationships and verifying existing ones within SNOMED CT. Our approach involves extracting and learning from the semantic context of SNOMED CT, an aspect that traditional frequency-based text analysis methods often struggle to capture. To overcome this weakness, we implement transformer architectures with BERT and RoBERTa as pivotal models. In this work, we utilize these models to tackle identifying *is-a* relationship, evaluating their efficacy in the SNOMED CT. The rest of the paper is organized as follows. Section II discusses related work. Section III describes the design of our proposed system, Section IV presents the results, and Section V discusses and concludes the paper, including plans for future work.

II. Background and Related Work

SNOMED CT uses "concepts" (unique identifiers) and "description" or "synonym" (human language expressions) to represent medical concepts. Each concept is associated with one or more descriptions.

Concepts within SNOMED CT are interconnected through relationships. These relationships provide computer-readable definitions of the concepts, which enables searching, retrieving, reusing, and analyzing information in various ways. The version of SNOMED CT (March 2022, U.S. Edition)[16] that we work with comprises a total of 2,563,333 relationships, encompassing 446,652 distinct concepts across 78 unique relation types. Among these relation types, about 40% of them (1,066,266) is the *is-a* relation.

Fig. 1 is a concept-relation diagram that outlines a formal representation of the concept *Abscess of back, except buttock (disorder)*. This concept is defined as a subclass of *Abscess of back (disorder)* through the *is-a* relation. The diagram also shows two other attribute type relationships: *Associated morphology* and *Finding site*. The *Associated morphology* attribute relation points to the concept *Abscess (morphologic abnormality)*. The *Finding site* attribute relation signifies that the abscess is located in a *Structure of back of trunk and/or neck (body structure)*. In essence, the diagram encapsulates the definition and relationships of the *Abscess of back, except buttock (disorder)* class in a formal, structured manner within the ontology.

To specifically identify the missing *is-a* relation, Cui et al.[17] group introduces a method to extract non-lattice subgraphs from SNOMED CT's *is-a* hierarchy and analyze lexical attributes of concepts, including those inherited from ancestors within these subgraphs. By comparing lexical attribute subsets between concept pairs to the existing *is-a* relationship, they identify potential missing *is-a* relationship in the September 2017 release of SNOMED CT (US edition).

Abeysinghe et al.[18] develop a method to detect the missing *is-a* relationship among concepts, classifying them into positive instances (with hierarchical relationships) and negative instances (without hierarchy despite apparent containment). Their model, trained on linguistic, semantic, hierarchical, and logical features, including enriched lexical attributes and information from knowledge bases, accurately predicts and classifies the relationships between concept pairs. This approach provides a detailed analysis for mapping complex conceptual hierarchies, leveraging comprehensive insights into linguistic, structural, and logical aspects.

Zheng et al.[19] introduce a novel transformation-based auditing method method using the Unified Medical Language System (UMLS) to identify 39,359 potentially missing hierarchical *is-a* relationship in medical terminologies. By dissecting concept names into base and secondary noun chunks and generating broader candidates for these parts, they highlight potential missing links. Their technique, which shows precision rates of 86.5% for SNOMED CT and 63% for Gene Ontology, effectively reveals undocumented hierarchical relationships.

Hao et al.[20] introduce a substring replacement method to identify potential missing relationships in SNOMED CT. By utilizing lexical attributes and the existing *is-a* relationship, they examine the March 2022 release of SNOMED CT US Edition[16], resulting in the proposal of 3,228 potential missing relationships. Among the proposed results, 63 of them are validated and confirmed by the UMLS. And 89 missing *is-a* relationships are manually validated by domain experts, from a random sample of 100 cases. The approach effectively pinpoints missing the *is-a* relationship in SNOMED CT.

Traditional word embedding methods, such as Word2Vec

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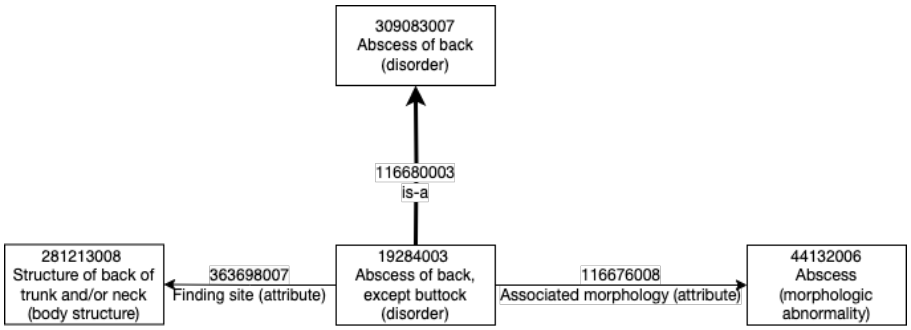

Fig. 1: An example of concepts and their relationship in SNOMED CT

and GloVe, capture semantic meaning by representing words as fixed vectors. However, these methods often fail to understand contextual information, as they assign the same representation to a word regardless of its meaning in different sentences. BERT undergoes pre-training on extensive text corpora and fine-tuning for specific applications, such as text classification and question answering. Its bidirectional architecture allows it to capture context from both directions in a sentence, making it exceptionally effective for tasks like semantic textual similarity (STS) and natural language inference (NLI).

BERT overcomes the limitations of traditional word embedding methods by offering robust contextual understanding, making it well-suited for diverse classification tasks. Fine-tuning BERT involves adding a classification layer on top of the pre-trained model and training it on a labeled dataset. In this research, we fine-tune BERT to generate embeddings for each pair and then classify the relationship as *is-a*. Our BERT model encodes each concept pairs into embeddings and identify the missing relation between undiscovered concepts.

III. Architecture and Methods

A. Baseline Method

The experiment establishes a baseline for comparing classical machine learning models (SVM, KNN, Naive Bayes, and MLP) by evaluating their performance in detecting SNOMED CT *is-a* relation. The dataset is tokenized using TF-IDF to create numerical feature vectors from the SNOMED CT terms and relations, which are then split into training and test sets. Each model is trained and evaluated using the same feature set, ensuring consistency across the comparisons. The results from each model are returned in a structured format, facilitating easy comparison of their effectiveness in detecting hierarchical *is-a* relationships. While the baseline experiment primarily focuses on classical models, it also includes provisions for future integration of graph-based model like Node2Vec, allowing for a more comprehensive comparison as the experiment expands. The modular design of the experiment supports the easy addition of new models and ensures that all evaluations are performed uniformly, making the baseline comparison robust and extendable.

- SVM: Support Vector Machines are utilized in sentiment classification by identifying an optimal hyperplane that separates sentiment classes in a high-dimensional feature space. This is achieved using text feature representations such as TF-IDF vectors or embeddings.
- KNN: The k-Nearest Neighbors (KNN) algorithm classifies sentiment by comparing the feature vector of a text instance to its k-nearest neighbors in the feature space and assigning the sentiment class based on the majority vote among those neighbors.
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B. Static Embedding Method

In order to determine if the *is-a* is the relation between concept pairs, we propose a framework based on previous studies of lexical features learning and relation classifier models and apply it to the entire SNOMED CT database. Specifically, we design a deep learning bi-classification model for *is-a* identification, starting with the original data package from SNOMED CT (March 2022, U.S. Edition)[16]. Specifically, the primary dataset for this study is extracted from a comprehensive collection of terms and their relationships, focusing on identifying *is-a* relationship among them. Every pair of concepts is linked by a specified relation, while any pair of concepts lacking a pre-established relation through SNOMED CT are regarded as instances without labels and will not be included in the training or testing data pool. We extract the input data from the description file and relation file respectively with the following data fields:

- *sourceId and destinationId*: unique identifiers for the source and destination terms, respectively.
- *relationGroup and typeId*: categorical variables indicating the group and type of the relation.
- *source_term and destination_term*: the textual representations of the source and destination entities involved in the relation.

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- *relation_term*: descriptive label of the relation between the source and destination terms.

TABLE I: An example of a pair of related concepts and their descriptions

Feature Names	Feature Values
sourceId	100000000
destinationId	102272007
typeId	116680003
source_term	BITTER-3
destination_term	Veterinary proprietary drug AND/OR biological
relation_term	*is-a*

This dataset serves as the foundation for exploring the classification of *is-a* relationship, which is pivotal for semantic analysis in various domains. The columns of *sourceId*, *destinationId*, and *typeId* are first extracted from the relation file Their textural description corpus then can be queried from the description file using each *conceptId*. A positive instance contains relationships of *is-a* and a negative instance contains concept pairs linked by relation types other than *is-a*. Table I shows an example of an extracted pair of concepts. In this case, it forms a positive instance as:

$$BITTER-3,$$
$$is-a,$$
$$Veterinary\ proprietary\ drug\ AND/OR\ biological$$

A negative example that contains both source and destination term with a relation type of the other 77 relation types in the SNOMED CT would be as:

$$atypical\ appendicitis\ (disorder),$$
$$associated\ morphology,$$
$$inflammation$$

We utilized the Pandas library in Python to load and preprocess the data. Each data instance consists of a source term, a binary indicator of the *is-a* relation class, and a destination term. The binary relation indicator categorizes the *is-a* relation as positive and all other relations as negative. To ensure a balanced representation, we applied the undersampling technique to obtain the quantity of each relation class up to 50,000 instances. In cases where a relationship has fewer than 10 instances, at least one instance is selected to preserve rare relationship types in the dataset.

The dataset is randomly divided into 80% for training and 20% for validation, ensuring effective model training and fine-tuning. The testing dataset, however, is an independent, unseen subset comprising revisions, modifications, and updates from future versions of the dataset. This methodology provides a comprehensive evaluation of the model's ability to generalize and accurately predict outcomes on updated data. For instance, if the model is trained on the 2018 version of the dataset, 80% of the 2018 data is used for training while 20% is allocated for validation. Testing is then performed on the 2019 version, which includes changes made between 2018 and 2019. This approach evaluates the model's robustness and adaptability to future updates and changes in the dataset.

After splitting the testing dataset, we conducted a post-split analysis to verify that the distributions of relationship types in the training and testing datasets are consistent with the original dataset. This analysis confirmed that both datasets accurately reflected the relationship type distribution of the original dataset, thereby ensuring the robustness and fairness of subsequent model training and evaluation phases.

The next step is tokenization, which transforms textual data into a sequence of tokens. We used the basic_English tokenizer from the TorchText library [21], which simplifies the text to a basic form suitable for processing. Following tokenization, a vocabulary was constructed from the training dataset's tokens, including a special token `<unk>` for unknown words. This vocabulary facilitates the conversion of text data into numerical form, enabling the neural network to process it.

The *embedding layer* transforms token indices into fixed size vectors, denoted as `embed_dim`. This transformation is pivotal for capturing the semantic essence of words within the model. Following the embedding process, the model employs a *maximum operation* to blend the embeddings derived from both the source and destination terms. This step is critical for retaining the most salient information from each input without resorting to concatenation or the use of attention mechanisms. The culmination of this process is the forwarding of the resultant vector through a *linear layer*. This layer functions to map the vector onto the designated output classes, effectively finalizing the classification task.

$$\text{Output} = \text{LinearLayer}\left(\max\left(\text{Embedding}(\text{source}),\right.\right.$$
$$\left.\left.\text{Embedding}(\text{destination})\right)\right)$$

Here, $\max(\cdot)$ denotes the maximum operation that combines the embeddings of the source and destination, ensuring that the most significant features are carried forward. The $\text{Embedding}(\cdot)$ function translates token indices into dense vectors, which are subsequently processed by the $\text{LinearLayer}(\cdot)$ to produce the final classification output.

The maximum operation was chosen because it achieved the highest accuracy compared to other operations (e.g., sum, mean). The performance for different concatenation methods is shown in Figure 3. The blue line (representing EmbedBagMax) consistently outperforms the other methods (EmbedBagMean and EmbedBagSum) in terms of accuracy. By the end of training, EmbedBagMax achieves accuracy close to 1.0, indicating its superior ability to capture key features and contribute effectively to the models predictions. This demonstrates that the max operation effectively highlights the most important feature(s) in the input, making it a robust aggregation method in the model.

In our approach, we utilize the information from the source concept and destination concept to create a featured representation of the encoded input data. To achieve this, we assign unique encodings to each source concept, destination concept, and relation concept. For example, consider the case of "Veterinary proprietary drug AND/OR biological" as the term, we first apply tokenization which extracts each basic

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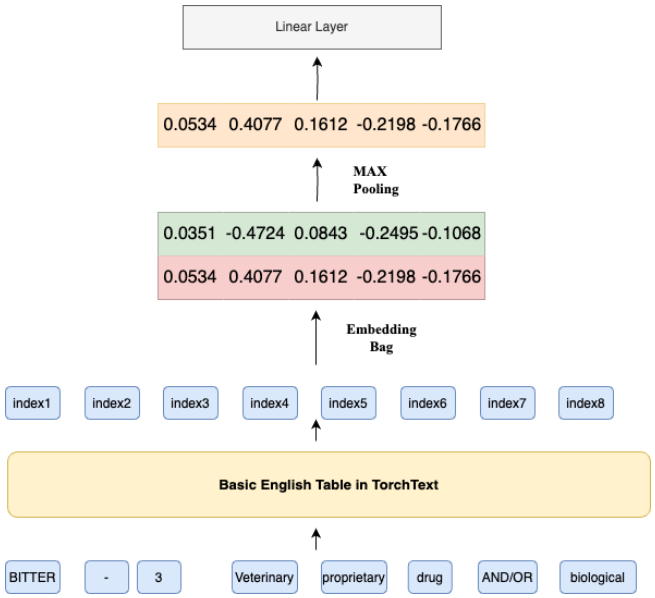

Fig. 2: Static embedding method

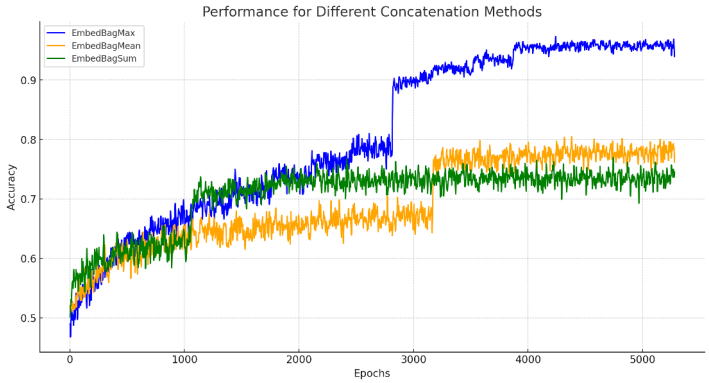

Fig. 3: Max operation performance the highest accuracy

English word as a token: Then map each token into an integer numerical index:

$$Veterinary : 1,$$
$$proprietary : 4,$$
$$drug : 3,$$
$$AND/OR : 0,$$
$$biological : 2$$

The process applies to both the relation term and the source term. Thus, the phrase "BITTER 3", "is a", "Veterinary proprietary drug AND/OR biological" can be transformed into [1, 4], [3, 0, 2], [1, 4, 3, 0, 2] in vector form, with lengths of 2, 3, and 5 respectively. Each instance will have its own size of vector representation. In deep learning, a neural network normally requires a fixed-size vector for input and output. To achieve this, we use EmbeddingBag[13], which converts vectors of varying lengths into a fixed-dimension vector, an example shown in Table II. This approach is also utilized in the work by Steinberg et al. [22] on preprocessing Electronic Health Record(EHR) datasets. The embedding layer effectively transforms descriptive text data, often consisting of words or indices, into dense vectors of fixed size (embed_dim), where each vector encapsulates the semantic meaning of its corresponding word within the textual context.

TABLE II: Representation of input indices, tokens, and embeddings

Tokens (from Vocabulary) **Embeddings (from nn.EmbeddingBag)**
['Veterinary_1', 'proprietary_4', 'drug_3', 'AND/OR_0', 'biological_2'] **[[0.0351, -0.4724, 0.0843, -0.2495, -0.1068]]**
['BITTER', '-_0', '3_2'] **[[0.0534, 0.4077, 0.1612, -0.2198, -0.1766]]**
['is_1', 'a_4'] **[[0.2007, -0.0163, -0.4048, 0.0722, 0.1369]]**

After the token embedding process, we employ a maximum

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embedding layer to sort the maximum value in each position of the [source concept], and [destination] pairs before inputting the data into the neural network, an example shown in Table III. By retaining the maximum value, the model emphasizes the dominant signal of the *is-a* concept pairs.

TABLE III: Max embedding of tensors

Concept 1 Embeddings	0.0351	-0.4724	0.0843	-0.2495	**-0.1068**
Concept 2 Embeddings	**0.0534**	**0.4077**	**0.1612**	**-0.2198**	-0.1766
Maximum Tensor	**0.0534**	**0.4077**	**0.1612**	**-0.2198**	**-0.1068**

After all the aforementioned processing steps, we train our model using stochastic gradient descent (SGD) with a learning rate specified by `lr`. A cross-entropy loss function is used to compute the difference between predicted and actual labels, guiding the model's learning. We implement a learning rate scheduler that reduces the learning rate based on the validation accuracy, allowing for finer adjustments as training progresses. The training process is divided into epochs, where each epoch involves a complete pass over the training data. During training, the model's parameters are adjusted to minimize the loss function. We use gradient clipping to prevent the exploding gradient problem, ensuring stable training. In our experiment, the gradient clipping parameter is set to 0.1 to ensure that the total gradient norm does not exceed this threshold. The gradient norm is computed using the L2 norm, and if it surpasses the specified limit, all gradients are scaled proportionally to reduce the total norm to 0.1. The model's performance is evaluated on a validation set, and the accuracy is reported to track progress.

After training, the model is evaluated on a separate test dataset to assess its generalization capability. This evaluation step is crucial for understanding the model's effectiveness in recognizing *is-a* relationship in unseen data.

The implementation leverages popular Python libraries such as Pandas for data manipulation, Torch and TorchText for model construction and training, and NLTK for natural language processing tasks like tokenization and stopword removal. The methodology emphasizes simplicity and effectiveness, aiming to develop a robust model for *is-a* relation class prediction tasks without the complexity of attention mechanisms or concatenated input representations.

C. LLM Pretrained Encoder Method

Embeddings and BERT encoders serve different purposes in natural language processing. Traditional embeddings like Word2Vec and GloVe provide static vector representations for words, which remain constant regardless of the context in which they appear. In contrast, BERT generates dynamic, context-sensitive embeddings that change based on the surrounding text, capturing nuanced meanings. This transition from static to contextual embeddings has significantly enhanced NLP capabilities. In the second part of this study, we apply this dynamic embedding encoder to test out the performance. Specifically, we explore both BERT and RoBERTa. BERT's context-sensitive embeddings allow for a deeper understanding of text, while RoBERTa refines the BERT model with optimized pre-training procedures to enhance performance. By leveraging these advanced encoders, we aim to achieve more accurate results in our NLP tasks, particularly in learning SNOMED CT concepts and relationships, where the advantage of context sensitivity in BERT and RoBERTa can lead to better feature learning and understanding.

We utilized historical data from the SNOMED CT US edition, spanning from 2017 to 2024. We trained our model on the data from 2017 and tested it on the incremental changes added in each subsequent year. This strategy allowed us to assess the model's performance based on the knowledge of concepts and relations from 2017 and evaluate its ability to predict unseen relations introduced in the following years.

To achieve this, we implemented a comprehensive data preprocessing and tokenization pipeline as follows:

Loading the Dataset: We load our datasets from CSV files which are specified for training, validation, and testing, and are read with a tab delimiter to parse the data correctly. **Tokenization**: The tokenization process is a critical preprocessing step where we convert textual data into a format suitable for the BERT/RoBERTa models. We processes the input sequences which is formatted by concatenating the source term, destination term with special tokens added to indicate the start, separation, and end of the sequence. Specifically, we used [CLS] to mark the beginning, [SEP] to separate concepts, and [END] to indicate the end of the sequence. To illustrate the process, consider the example where the source concept is *BITTER 3*, and the destination concept is *Veterinary proprietary drug AND/OR biological*, resulting in a label of 1. The tokenization step involves converting these concepts into a format that the models can process.

For BERT, the tokenization produces the following tokens: [CLS], *BITTER*, -, *3*, [SEP], *Veterinary*, *proprietary*, *drug*, *AND*, */*, *OR*, *biological*, [SEP], [END]. Each token is then mapped to its corresponding ID in the model's vocabulary. The token IDs for this example is:

$$[101, 10712, 1011, 1017, 102, 19337, 18550, 4310, 2143$$
$$1015, 1013, 2030, 3178, 102, 2003, 1011, 1037, 1001]$$

An attention mask is also generated, where each token is assigned a 1 (to denote actual tokens) or a 0 (for padding tokens).

Handling Variable Length Sequences: To manage input sequences of different lengths, we employed padding and truncation. Padding ensures that all sequences reach a uniform length by adding special padding tokens where necessary. Truncation ensures that sequences exceeding the maximum allowable length of 120 are shortened to fit the model's input size requirements.

By following these preprocessing steps, we ensured that our data was properly formatted and ready for training

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BERT/RoBERTa models. This preprocessing pipeline facilitated consistent and effective input handling.

Training on the 2017 data and testing on updates from subsequent years allowed us to evaluate the model's ability to predict missing *is-a* relations and identify undefined *is-a* relation concept pairs across the SNOMED CT dataset. This method demonstrated the model's effectiveness in handling evolving data and maintaining accuracy and relevance over time. By systematically preparing the data in this way, we ensured a robust input pipeline that leverages the strengths of BERT/RoBERTa encoders for *is-a* relation classification prediction tasks.

Dynamic Embedding with BERT: Traditional word embeddings like Word2Vec and GloVe assign a fixed vector representation to each word, regardless of context. In contrast, BERT generates dynamic embeddings that adapt based on the surrounding text. This is crucial for understanding the contextual meaning of each word in the sequence.

For example, the word *drug* in the context of *Veterinary proprietary drug AND/OR biological* would receive a different embedding than *drug* in a different sentence due to the context provided by surrounding words.

Self-Attention Mechanism: The self-attention mechanism allows the model to weigh the importance of each token in the sequence relative to the others. This is crucial for understanding the relationships between the source and destination concepts, as well as the relation type.

During self-attention, each token in the input sequence is transformed into three vectors: the Query, Key, and Value vectors. These vectors help determine the importance of each token in the context of the entire sequence.

For example, the token *BITTER* has the following transformations:

- Query vector: Represents the current token's significance.
- Key vector: Represents the context in which the token should be considered.
- Value vector: Represents the actual information of the token.

The self-attention mechanism calculates the attention scores by taking the dot product of the Query vector of one token with the Key vectors of all tokens, followed by applying a softmax function to obtain the attention weights. These weights determine how much focus each token should receive.

In our example, if the Query vector for *BITTER* and the Key vector for *Veterinary* produce a high attention score, it indicates a strong relationship between these tokens. The weighted sum of the Value vectors, using the attention weights, produces the final representation of the token.

Forward Pass: In the forward pass, the token IDs and attention masks are passed through the embedding layer of the model. This converts them into dense vectors that encapsulate the semantic meaning of the tokens. For instance, the token ID [10712] corresponding to *BITTER* is mapped to an embedding vector:

```
[-0.05, 0.42, -0.54, ... , 0.12]
```

These embeddings are processed through multiple layers of self-attention and feed-forward networks. Each self-attention layer allows the model to weigh the importance of each token in the sequence, effectively capturing the relationship between the source and destination concepts as well as the relation type.

The final output layer produces a vector representing the probability distribution over the classes (0 or 1). For our example, the model outputs a probability for class 1, indicating the *is-a* relationship.

Training Process: The training process involves feeding the tokenized and embedded input through the model, computing the loss, and using backpropagation to adjust the model parameters. The optimizer updates the model parameters to minimize the loss iteratively.

Validation and Testing: After training, the model's performance is evaluated on validation and test datasets. For each input pair, the model generates predictions and these are compared against the true labels. In our example, if the true label is 1 and the model predicts a high probability for class 1, this indicates good performance.

Performance metrics such as accuracy, precision, recall, F1 score, and ROC AUC are calculated to assess the model. For instance, if the model correctly predicts 95 out of 100 *is-a* relationships, its accuracy is 95%.

In this study, BERT and RoBERTa models were utilized to classify relationships within SNOMED CT concept pairs. Using the example where the source concept is *BITTER 3* and the destination concept is *Veterinary proprietary drug AND/OR biological* with the relation *is-a* the process involved tokenization, input formatting, and embedding through multiple layers of self-attention in the model. The training phase included loss calculation, backpropagation, and optimization to enhance the model's ability to correctly classify the *is-a* relationship. The model's performance was evaluated through standard metrics to ensure its robustness and accuracy in predicting complex medical relationships.

BERT and RoBERTa for SNOMED CT. RoBERTa is pre-trained on a significantly larger and more diverse dataset compared to BERT, which potentially enables it to capture a broader range of language patterns. Additionally, RoBERTa omits the Next Sentence Prediction (NSP) task and employs dynamic masking, which may enhance its generalization capabilities and overall performance when applied to the SNOMED CT dataset.

RoBERTa employs larger batch sizes and longer training durations, which may contribute to its superior fine-tuning performance on the SNOMED CT dataset. The optimization techniques and training strategies incorporated in RoBERTa are specifically designed to address the limitations identified in BERT, further improving its effectiveness.

Despite both BERT and RoBERTa utilizing the transformer architecture and self-attention mechanisms, the rigorous training regimen of RoBERTa is likely to produce more robust contextual embeddings. This robustness can enhance the classification accuracy for complex relationships within the SNOMED CT dataset.

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Both models undergo a similar fine-tuning process on the SNOMED CT dataset, where their pre-trained weights are adjusted to better predict *is-a* relationships. However, the enhancements in RoBERTa's pre-training phase might result in more effective fine-tuning outcomes, leading to improved performance in identifying these relationships.

IV. Results

A. Static Embedding Result

During the training phase, we started with a learning rate of 20 over 15 epochs, reducing the learning rate to 10% of its previous value after each epoch. With a batch size of 326, the model's training accuracy improved rapidly and then stabilized around 95% after about 1,000 batches, indicating a quick learning followed by convergence (Figure 4). The high learning rate was found to be effective in quickly converging to a good solution, reducing training time, and improving model efficiency. In this study, we evaluated the performance of our text classification model in identifying *is a* relationships within a balanced binary dataset. The model's effectiveness was quantified using precision, recall, and F1 score metrics, along with a detailed analysis provided by a confusion matrix. Table IV shows that the model achieved a precision of 90.31%, indicating a high rate of relevant instances among the retrieved ones. The recall score was slightly lower at 89.46%, reflecting the model's ability to identify most of the relevant instances. The balanced measure of precision and recall, the F1 score, is 89.88%, showcasing the model's robust performance in balancing both metrics. Table V shows that the model accurately identified 4,509 instances as not having an *is-a* relationship and 4,482 instances as having one. It incorrectly labeled 481 instances as having an *is-a* relationship and overlooked 528 instances that did. According to the confusion matrix, the results demonstrate the model's strong capability in accurately identifying *is-a* relationship within the dataset, achieving high precision and recall. The relatively low false positive and false negative rates, as depicted in the confusion matrix, further attest to the model's efficacy. These outcomes highlight the model's potential in semantic relation classification tasks, particularly in applications where identifying *is-a* relationship is crucial.

TABLE IV: Result measurement

Metric	Score
Precision	**0.903**
Recall	**0.895**
F1 Score	**0.899**

TABLE V: Confusion matrix

	Predicted Negative	Predicted Positive
Actual Negative	4509	481
Actual Positive	528	4482

B. LLM Pretrained Encoder Result

The results of the performance comparison between the BERT and RoBERTa models are presented in Table VI. Both models were evaluated on the same dataset of newly released data for each year, using various performance metrics, including accuracy, precision, recall, F1 score, and ROC AUC. The test data from the 2021 edition yielded the highest evaluation scores.

As a result of new released data in year 2024, **the accuracy** of the RoBERTa model 0.926 slightly surpasses that of the BERT model 0.918. This indicates that RoBERTa correctly classified a marginally higher proportion of instances compared to BERT. **Precision**, which measures the correctness of positive predictions, is also higher for RoBERTa 0.926 than for BERT 0.918, suggesting that RoBERTa has a better capability in identifying relevant instances among the predicted positives. **The recall** of the RoBERTa model 0.926 exceeds that of the BERT model 0.918, reflecting that RoBERTa is slightly better at identifying all relevant instances in the dataset. **The F1 score**, which balances precision and recall, is higher for RoBERTa 0.9259 compared to BERT 0.9179, further emphasizing RoBERTa's superior performance in achieving this balance. Both models achieved a perfect **ROC AUC** score of 0.971, indicating their excellent performance in distinguishing between the positive and negative classes. The total running time of this experiment was approximately 13 hours on a single V100 GPU unit.

TABLE VI: Test results with yearly updated dataset
BERT & RoBERTa
(BERT: learning rate=2×10^{-5}, epochs=30, batch=100;
RoBERTa: learning rate=2×10^{-5}, epochs=30, batch=100)

Test	Models	Recall	Precision	F1	Accuracy	ROC
2024	BERT	0.918	0.918	0.918	0.918	0.971
	RoBERTa	**0.926**	**0.926**	**0.925**	**0.926**	0.971
2023	BERT	0.894	0.895	0.893	0.894	**0.957**
	RoBERTa	**0.897**	**0.904**	**0.896**	**0.897**	0.950
2022	BERT	0.899	0.900	0.890	0.899	0.950
	RoBERTa	**0.912**	**0.912**	**0.911**	**0.912**	**0.961**
2021	BERT	0.892	0.899	0.892	0.892	0.956
	RoBERTa	**0.933**	**0.933**	**0.932**	**0.933**	**0.972**

In summary, we used BERT and RoBERTs to train on the 2017 edition of SNOMED CT with the same tuning parameters, including a learning rate of 2×10^{-5} and 30 epochs. Both models demonstrated exceptional performance in the pair concept relation classification tasks, achieving high accuracy, precision, recall, and F1 score. Between the two models, RoBERTa has a slight edge.

V. Discussion and Conclusion

A. Discussion

We compare the baseline performance of the models across three main groups: frequency-based classic model, graph-based models, and a transformer-based approach (RoBERTa). These groups represent different paradigms in machine learning and natural language processing, allowing for a more comprehensive analysis of their strengths and weaknesses in addressing the classification task. The table VII provides a comparison of several models (SVM, KNN, Naive Bayes,

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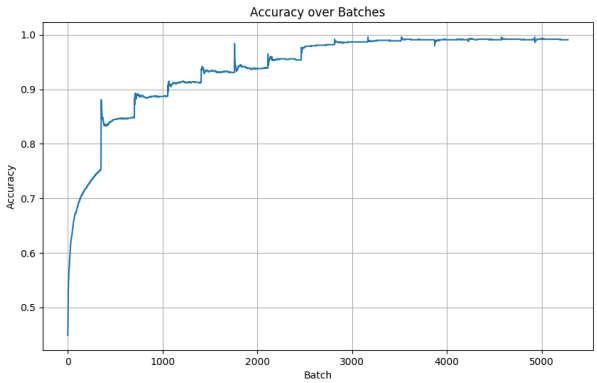

Fig. 4: The training process

TABLE VII: frequency-based vs graph-based vs transformer-based performance comparison

Models	Recall	Precision	F1	Accuracy	ROC
SVM	0.83	0.817	0.859	0.837	0.930
KNN	0.822	0.835	0.804	0.819	0.900
Naive Bayes	0.564	0.630	0.321	0.426	0.583
Node2Vec	0.689	0.675	0.736	0.704	0.747
RoBERTa	**0**.926	0.926	0.925	**0**.926	**0.971**

Node2Vec, and RoBERTa) based on five performance metrics: Recall, Precision, F1-Score, Accuracy, and ROC AUC. Among these models, RoBERTa stands out as the best-performing model in almost every category.

Frequency-Based Classic Models: These models rely on traditional, feature-based approaches where the relationships between features and classes are defined using mathematical functions and assumptions about data distributions. SVM is the top performer among the classic models, with an accuracy of 0.8371 and a ROC AUC of 0.930, indicating that it effectively balances recall and precision. Its high F1-Score (0.8587) shows that it performs well across both positive and negative classes. KNN demonstrates competitive results with a slightly lower accuracy 0.8191 and ROC AUC 0.8995 compared to SVM. Its F1-Score of 0.8040 reflects that it performs decently but suffers from slightly reduced generalization ability, likely due to the curse of dimensionality or suboptimal distance metrics in this feature space. Naive Bayes is the weakest of the classic models, with poor recall 0.564 and F1-Score 0.3214. This suggests that the model struggles with misclassification, likely due to its reliance on strong independence assumptions, which may not hold in this complex dataset.

Graph-Based Model: This model focuses on graph structure to capture the relationships between data points, such as connections between entities or features, by learning node embeddings from graph representations of the data. Node2Vec, with a F1-Score of 0.704 and a ROC AUC of 0.747, but it still lags behind the frequency-based models like SVM. While Node2Vec captures some relationships between graph nodes, it struggles with precision and recall compared to transformer-based models, likely due to its limitation in handling non-linear relationships in the data.

Transformer-Based Approach: RoBERTa is a transformer-based model pre-trained on large corpora and fine-tuned for specific tasks. It uses self-attention mechanisms to capture complex relationships within the data, learning both syntactic and semantic patterns that other models might miss. RoBERTa achieves the highest performance across almost all metrics, with an accuracy of 0.926, a precision of 0.926, and a ROC AUC of 0.971. Its F1-Score of 0.925 shows a balanced handling of both precision and recall, indicating minimal trade-offs between capturing true positives and avoiding false positives. The performance boost provided by RoBERTa is likely due to its ability to capture deep contextual relationships within the data, thanks to its transformer architecture and the massive amount of pre-training on diverse text corpora.

Overall, RoBERTa is the clear winner among the compared methods, setting a high standard for model performance in this task. While traditional models like SVM are still competitive, and graph-based models like Node2Vec show some promise, the superior performance of RoBERTa highlights the advantages of modern transformer-based models in capturing intricate data relationships that other models may miss.

The classification tasks' results indicate that BERT and RoBERTa perform superiorly against the static embedding method across all reported metrics. The static embedding method, which trains on the most recently released data, achieved an F1 score of 0.899. In contrast, despite being trained on the 2017 edition of SNOMED CT, the dynamic embedding methods demonstrated marked improvements, with BERT and RoBERTa attaining F1 scores of 0.918 and 0.925, respectively, on the 2024 edition of SNOMED CT. Both methods are evaluated using the newly released 2024 edition of SNOMED CT. These findings underscore the efficacy of transformer-based learning in providing enhanced feature representations, which contribute to the classification models' overall performance. Notably, the superior performance of dy-

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namic embeddings, even when trained on older and potentially outdated data, highlights their robustness and adaptability in handling newer data.

The original dataset, focused on *is-a* relationship, might not be representative of all semantic relationships present in natural language, leading to a selection bias. This limitation could affect the model's generalizability to broader types of relationship tasks. Simplifying the dataset to a binary classification task (*is-a* vs. *non-is-a*) might overlook the nuances and complexities inherent in semantic relationships. Some relationships might not be strictly binary or interpreted differently in various contexts. Future work could focus on improving model performance by incorporating knowledge graph network representation, a richer feature set for deeper learning. Expanding the model's use across diverse datasets and languages can also enhance its flexibility and effectiveness.

The unique contribution of this study is its exclusive reliance on the SNOMED CT system as the sole data source. Unlike previous research incorporating additional ontology databases to annotate or enrich information when relation pair data is sparse, this study operates entirely within the constraints of SNOMED CT. Additionally, it explores the application of pre-trained transformer models, a novel approach that has not been extensively utilized in prior studies involving this dataset or similar medical ontologies.

B. Conclusion

This research focuses on developing a concept-pair classification model to identify *is-a* relationship in the SNOMED CT ontology dataset. Our model demonstrated remarkable performance metrics, achieving a precision of 93.3%, a recall of 93.3%, and an F1 score of 93.2%. These results show that our method can accurately and reliably discern *is-a* relationship, and to enhance accuracy and coverage for SNOME CT ontology development. In future work, we aim to extend our approach to identify multiple relationship types within SNOMED CT, enabling a more comprehensive and detailed compilation of relationships.

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References

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." arXiv, May 24, 2019. doi: 10.48550/arXiv.1810.04805.

[2] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach." arXiv, Jul. 26, 2019. doi: 10.48550/arXiv.1907.11692.

[3] E. Chang and J. Mostafa, "The use of SNOMED CT, 2013-2020: a literature review," J Am Med Inform Assoc, vol. 28, no. 9, pp. 2017–2026, Aug. 2021, doi: 10.1093/jamia/ocab084.

[4] P. I. Dissanayake, T. K. Colicchio, and J. J. Cimino, "Using clinical reasoning ontologies to make smarter clinical decision support systems: a systematic review and data synthesis," Journal of the American Medical Informatics Association, vol. 27, no. 1, pp. 159–174, Jan. 2020, doi: 10.1093/jamia/ocz169.

[5] "Practical Guide to Reference Sets - Reference Sets Practical Guide - SNOMED Confluence." https://confluence.ihtsdotools.org/display/DOCRFSPGpreview=/441\protect\@normalcr\relax07986/45529216/doc_ReferenceSetPracticalGuide_Current-en-US_INT_20170516.pdf(accessed Aug.03,2023).

[6] C. Gaudet-Blavignac, V. Foufi, M. Bjelogrlic, and C. Lovis, "Use of the Systematized Nomenclature of Medicine Clinical Terms (SNOMED CT) for Processing Free Text in Health Care: Systematic Scoping Review," J Med Internet Res, vol. 23, no. 1, p. e24594, Jan. 2021, doi: 10.2196/24594.

[7] P. L. Elkin et al., "Evaluation of the content coverage of SNOMED CT: ability of SNOMED clinical terms to represent clinical problem lists," Mayo Clin Proc, vol. 81, no. 6, pp. 741–748, Jun. 2006, doi: 10.4065/81.6.741.

[8] K. Donnelly, "SNOMED-CT: The advanced terminology and coding system for eHealth," Stud Health Technol Inform, vol. 121, pp. 279–290, 2006.

[9] G. B. Melton, S. Parsons, F. P. Morrison, A. S. Rothschild, M. Markatou, and G. Hripcsak, "Inter-patient distance metrics using SNOMED CT defining relationships," *Journal of Biomedical Informatics*, vol. 39, no. 6, pp. 697–705, Dec. 2006, doi: 10.1016/j.jbi.2006.01.004.

[10] R. Cornet and N. de Keizer, "Forty years of SNOMED: a literature review," *BMC Med Inform Decis Mak*, vol. 8, no. 1, p. S2, Oct. 2008, doi: 10.1186/1472-6947-8-S1-S2.

[11] İ. Karadeniz and A. Özgür, "Linking entities through an ontology using word embeddings and syntactic re-ranking," BMC Bioinformatics, vol. 20, no. 1, p. 156, Mar. 2019, doi: 10.1186/s12859-019-2678-8.

[12] M. Mohtashamian, R. Abeysinghe, X. Hao, and L. Cui, "Identifying Missing IS-A Relations in Orphanet Rare Disease Ontology," in 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Dec. 2022, pp. 3274–3279. doi: 10.1109/BIBM55620.2022.9995614.

[13] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," in Advances in Neural Information Processing Systems, Curran Associates, Inc., 2019. Accessed: Jun. 11, 2024. [Online]. Available: https://papers.neurips.cc/paper_files/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html

[14] S. Singhal, B. Hegde, P. Karmalkar, J. Muhith, and H. Gurulingappa, "Weakly Supervised Learning for Categorization of Medical Inquiries for Customer Service Effectiveness," Front Res Metr Anal, vol. 6, p. 683400, 2021, doi: 10.3389/frma.2021.683400.

[15] X. Yang, D. Lo, X. Xia, and J. Sun, "Condensing Class Diagrams With Minimal Manual Labeling Cost," in 2016 IEEE 40th Annual Computer Software and Applications Conference (COMPSAC), Jun. 2016, pp. 22–31. doi: 10.1109/COMPSAC.2016.83.

[16] "SNOMED CT United States Edition." Available: https://www.nlm.nih.gov/healthit/snomedct/us{_}edition.html.[Accessed:Aug.03,2023]

[17] L. Cui, O. Bodenreider, J. Shi, and G.-Q. Zhang, "Auditing SNOMED CT hierarchical relations based on lexical features of concepts in non-lattice subgraphs," Journal of Biomedical Informatics, vol. 78, pp. 177–184, Feb. 2018, doi: 10.1016/j.jbi.2017.12.010.

[18] R. Abeysinghe, F. Zheng, E. V. Bernstam, J. Shi, O. Bodenreider, and L. Cui, "A deep learning approach to identify missing is-a relations in SNOMED CT," Journal of the American Medical Informatics Association, vol. 30, no. 3, pp. 475–484, Mar. 2023, doi: 10.1093/jamia/ocac248.

[19] F. Zheng, J. Shi, Y. Yang, W. J. Zheng, and L. Cui, "A transformation-based method for auditing the IS-A hierarchy of biomedical terminologies in the Unified Medical Language System," Journal of the American Medical Informatics Association, vol. 27, no. 10, pp. 1568–1575, Oct. 2020, doi: 10.1093/jamia/ocaa123.

[20] X. Hao, R. Abeysinghe, J. Shi, and L. Cui, "A substring replacement approach for identifying missing IS-A relations in SNOMED CT," in 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), Dec. 2022, pp. 2611–2618. doi: 10.1109/BIBM55620.2022.9995595.

[21] "Language Translation with TorchText — PyTorch Tutorials 1.7.1 documentation." Accessed: Jun. 11, 2024. [Online]. Available: https://pytorch.org/tutorials/beginner/torchtext_translation_tutorial.html

[22] E. Steinberg, K. Jung, J. A. Fries, C. K. Corbin, S. R. Pfohl, and N. H. Shah, "Language models are an effective representation learning technique for electronic health record data," Journal of Biomedical Informatics, vol. 113, p. 103637, Jan. 2021, doi: 10.1016/j.jbi.2020.103637.

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